# KASHISH KHAN

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## PROFESSIONAL SUMMARY

MSc Artificial Intelligence graduate (Dublin Business School, 2026) with end-to-end experience building LLM and RAG systems: a self-hosted LLM observability platform deployed to Azure Container Apps; a five-layer GraphRAG clinical decision support pipeline over 50,000+ MIMIC-IV records (89.7% section hit rate; BERTScore F1 = 0.77; ROUGE-1 = 0.61 on structured clinical tasks); and a hackathon-winning AI inbox triage system (Give(A)Go × Lette AI, 2026). Strong across the full ML stack: BioClinicalBERT, FAISS, Neo4j, FastAPI, Docker. Targeting junior AI/NLP/ML engineering and data science roles in the Irish market.

## KEY PROJECTS

**GraphRAG Clinical Decision Support** *2025–2026 | MSc Dissertation | NLP / KG / RAG*

*Python, BioClinicalBERT, FAISS, Neo4j, MongoDB, LangChain, HuggingFace, Docker, spaCy (SciSpaCy)*

- Engineered a five-layer RAG pipeline over 50,000+ MIMIC-IV discharge summaries: BioClinicalBERT 768-dim FAISS dense retrieval, BM25 lexical reranking, Neo4j KG concept overlap, section bias, and child-span refinement combined into a single hybrid scoring formula.
- Child-span refinement re-embeds parent chunks into sentence-level sub-spans and re-ranks by cosine similarity — sentence-level precision from a chunk-level index with no storage overhead.
- **Benchmark (50 tasks / 107 cases):** Hit@5 = 1.0; section hit rate 89.7%; gold chunk retrieval 87.9%; Precision@5 = 45.2%; BERTScore F1 = 0.77, ROUGE-1 = 0.61 (pmh_list); BERTScore F1 = 0.51 (chief complaint).
- Section-aware chunking (7 types; 240T narrative / 420T list / 160T header targets) with deterministic content-hash chunk IDs and idempotent processing across re-indexing cycles.

**LLM Trace & Cost Studio** *2025–2026 | Personal Build | Azure / Docker / FastAPI*

*Python, FastAPI, Streamlit, Docker Compose, Azure Container Apps, LangChain, OpenAI API, SQLite*

- Designed and shipped a self-hosted LLM observability platform (LangSmith-lite): async FastAPI backend, Streamlit analytics dashboard, SQLite trace store — containerised and deployed to Azure Container Apps.
- Implemented per-request latency tracking, token-usage aggregation, and cost attribution across OpenAI API calls; deterministic chunk-ID mapping detects embedding drift across re-indexing cycles.
- Applied production engineering patterns: centralised config, idempotent guards, type annotations, graceful fallback chains throughout.

**Lette Inbox Copilot — Hackathon Winner** *2026 | Give(A)Go × Lette AI*

*Python, Streamlit, OpenAI API (GPT-4o-mini) — github.com/KashishhKhann/Lette*

- Won the Lette AI track at Give(A)Go 2026; built a three-tier property-management inbox triage system: auto-resolve, AI-draft (GPT-4o-mini), and human-escalation tier for low-confidence queries.
- Real-time Streamlit dashboard with confidence scores and tier routing; received direct outreach from Lette co-founder following the event.

## EDUCATION

**MSc Artificial Intelligence** — Dublin Business School *Sept 2025 – June 2026 | Dublin*

- **Modules:** Advanced NLP, Deep Reinforcement Learning, Cognitive & Ethical AI, Cloud Computing.
- Implemented MiniBioClinicalBERT from scratch in Julia (Flux.jl): positional encodings, multi-head self-attention, transformer blocks, cross-validation — one of few MSc projects building a transformer end-to-end in a non-Python ML framework.
- Custom Gymnasium RL environment (RealisticEREnv + Pygame); benchmarked Dueling DQN vs Baseline DQN.

**MCA — Master of Computer Applications** — Amity University *2022–2024 | India*

- Built clinical NLP pipeline (Scrapy, NLTK, MongoDB, PostgreSQL): news sentiment, media bias detection, PySpark CRISP-DM healthcare pipeline with smoker classification.
- Delivered cross-platform voice assistant (SQLite backend, GUI) and graph centrality analysis system (Neo4j, NetworkX).

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, Julia, SQL, Bash |
| **LLM / GenAI** | LangChain, LlamaIndex, OpenAI API, HuggingFace Transformers, RAG Pipelines, Prompt Engineering, LoRA/PEFT |
| **NLP** | BERT, BioClinicalBERT, SciSpaCy, Sentence Transformers, Med7, NLTK |
| **ML / DL** | PyTorch, Scikit-learn, Flux.jl, Lux.jl |
| **Vector / KG** | FAISS (IndexFlatIP, HNSW), ChromaDB, Neo4j (Cypher), NetworkX |
| **Databases** | MongoDB, PostgreSQL, SQLite |
| **MLOps / Infra** | Docker, FastAPI, Streamlit, Azure Container Apps, GCP, MLflow, pytest |
| **Data Eng.** | Pandas, NumPy, Scrapy, PySpark, Matplotlib, Seaborn |
| **Dev Tools** | Git, GitHub, VS Code, Jupyter, CI/CD (GitHub Actions) |

## ACHIEVEMENTS

- **Hackathon Winner** — Give(A)Go × Lette AI (2026): selected as winning entry; received direct post-event outreach from co-founder.
- **Dual Master's Degrees** — MSc AI (2026, Dublin Business School) + MCA (2024, Amity University).
- Built MiniBioClinicalBERT from scratch in Julia (Flux.jl): positional encodings, multi-head self-attention, transformer blocks, cross-validation.
- **Google Cloud Computing Foundations Certificate** (Dec 2023).